House Price Prediction Using Machine Learning

Project Summary

Objective

This project's goal was to create a machine learning model that could forecast home values based on a variety of property attributes, including area, number of bedrooms, bathrooms, stories, parking facilities, furnishing status, and location-related aspects. The project's objectives were to examine the effectiveness of several regression models and determine the elements that most significantly affect home prices.

Dataset Description

The Housing dataset was used for this project. The dataset contains information about residential properties, including both numerical and categorical features. The target variable for prediction was the house price. Data exploration was performed to understand the structure of the dataset, identify missing values, and analyze relationships between variables.

Data Preprocessing

The dataset was cleaned and prepared for machine learning by checking for missing values and duplicate records. Categorical features such as main road access, guest room availability, basement, air conditioning, preferred area, and furnishing status were converted into numerical format using one-hot encoding. This preprocessing ensured that the data was suitable for regression modeling.

Machine Learning Models

Two regression algorithms were put into practice and assessed:

Regression in Linear Form

Regressor for Random Forests

An 80:20 ratio was used to split the dataset into training and testing sets. Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R2 Score were used to assess the model's performance.

Data Visualization

To learn more about the dataset, a number of visualizations were made, such as:

Price distribution for homes

Heatmap of correlation

Scatter plot of price versus area

Comparison between actual and anticipated prices

Important correlations between property attributes and home values were found thanks to these visualizations.

Key Findings

According to the data, one of the biggest factors affecting home prices is property area. The value of a house is also greatly influenced by other characteristics like the number of bathrooms, the location of choice, air conditioning, and the quality of the furnishings. In the market, larger homes typically fetch greater costs.

The Random Forest Regressor outperformed Linear Regression in terms of predictive performance among the investigated models, demonstrating its capacity to identify more intricate associations in the data.

Conclusion

This experiment effectively illustrated how machine learning methods may be used to forecast home prices. Important insights on the factors influencing property prices were discovered through data pretreatment, visualization, model building, and evaluation. The created models can help buyers, sellers, and real estate companies make better price selections. To further increase forecast accuracy, future developments might make use of bigger datasets and test more sophisticated machine learning methods.

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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Jupyter Notebook